Status	Finished
Started	Wednesday, 14 October 2026, 7:32 PM
Completed	Wednesday, 14 October 2026, 7:49 PM
Duration	17 mins 13 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

What type of problems can sorting help solve efficiently?

- a. Duplicates
- b. Searching
- c. Selection
- d. All of the above

The correct answer is: All of the above

Question **2**

Complete

Mark 1.00 out of 1.00

What is the primary benefit of using sorting algorithms in programming?

- a. Makes data harder to manage
- b. Provides a basis for other algorithms to work efficiently
- c. Decreases the efficiency of algorithms
- d. Makes code execution slower

The correct answer is: Provides a basis for other algorithms to work efficiently

Question **3**

Complete

Mark 1.00 out of 1.00

What is the primary advantage of the divide-and-conquer approach in sorting algorithms?

- a. It only works on small datasets
- b. It allows for efficient parallel processing and sorting of data
- c. It simplifies the sorting process by using only one pass
- d. It avoids the need for recursion

The correct answer is: It allows for efficient parallel processing and sorting of data

Question **4**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after the two halves of the list are sorted?

a. They are split again into smaller sublists
b. They are combined to form a single sorted list
c. They are discarded
d. They are compared element by element

The correct answer is: They are combined to form a single sorted list

Question **5**

Complete

Mark 1.00 out of 1.00

Why is it advantageous to sort data before performing duplicate analysis?

a. It has no effect on the analysis process
b. It allows for quicker identification of duplicates
c. It complicates the analysis process
d. It makes the analysis slower

The correct answer is: It allows for quicker identification of duplicates

Question **6**

Complete

Mark 1.00 out of 1.00

Algorithm design technique used in merge sort algorithm is

a. Divide and conquer
b. Greedy method
c. Dynamic programming
d. Backtracking

The correct answer is:
Divide and conquer

Question **7**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after dividing the input into smaller parts?

a. The parts are ignored
b. The parts are merged without sorting
c. Each part is sorted independently
d. Each part is searched for a specific element

The correct answer is: Each part is sorted independently

Question **8**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is known for its simplicity and sequential comparison of elements?

a. Merge Sort
b. Bubble Sort
c. Quick Sort
d. Heap Sort

The correct answer is: Bubble Sort

Question **9**

Complete

Mark 1.00 out of 1.00

What does the Bubble Sort algorithm primarily focus on during each pass?

a. Dividing the list into halves
b. Bubbling up the smallest element
c. Sorting the entire list in one pass
d. Bubbling up the largest element to its correct position

The correct answer is: Bubbling up the largest element to its correct position

Question **10**

Complete

Mark 1.00 out of 1.00

How does Merge Sort achieve its efficiency?

- a. By breaking the input into smaller parts and merging them
- b. By comparing elements sequentially
- c. By using the bubble-up method
- d. By sorting data in a single pass

The correct answer is: By breaking the input into smaller parts and merging them

Question **11**

Complete

Mark 1.00 out of 1.00

Which of the following is a key reason for the importance of sorting algorithms?

- a. Sorting helps in finding duplicates quickly
- b. Sorting makes it harder to search for items
- c. Sorting is rarely used in programming
- d. Sorting decreases the efficiency of selection operations

The correct answer is: Sorting helps in finding duplicates quickly

Question **12**

Complete

Mark 1.00 out of 1.00

What is one advantage of sorting a list before performing a search operation?

- a. It makes the search operation slower
- b. It increases the number of comparisons needed
- c. It allows for faster searching
- d. It has no effect on the search operation

The correct answer is: It allows for faster searching

Question **13**

Complete

Mark 1.00 out of 1.00

What is sorting in the context of computer science?

a. Inserting data into a list
b. Deleting data from a list
c. Arranging data in a particular format
d. Searching for data in a list

The correct answer is: Arranging data in a particular format

Question **14**

Complete

Mark 1.00 out of 1.00

_____is putting an element in the appropriate place in a sorted list yields a larger sorted order list.

a. Distribution
b. Extraction
c. Insertion
d. Selection

The correct answer is:
Insertion

Question **15**

Complete

Mark 1.00 out of 1.00

In the context of sorting, what does the divide-and-conquer approach involve?

a. Sorting data sequentially
b. Dividing the input into parts, solving each part, and combining the solutions
c. Sorting data in a single pass
d. Rearranging data without sorting

The correct answer is: Dividing the input into parts, solving each part, and combining the solutions